Minh Nhat Dang

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PROFESSIONAL EXPERIENCE

Venera AI **New York, NY**
Senior Machine Learning & AI Engineer *Jun 2024 – Present*

- Led development of a bilingual medical AI system covering EHR-grounded data generation, SFT, evaluation, inference optimization, deployment, and monitoring
- Designed agentic QA workflows with tool-use control, retrieval/search integration, structured outputs, retry logic, and prompt-following safeguards for healthcare chatbot use cases
- Fine-tuned and benchmarked 0.6B–32B Qwen/Llama/Gemma models using Unsloth, LoRA/QLoRA, DeepSpeed ZeRO-2, LlamaFactory, vLLM, and LM-Eval
- Reduced GPU memory footprint by 40%, fine-tuning cost by 3x, and inference latency by 25% through quantization, batching, context tuning, and vLLM serving optimization
- Built evaluation infrastructure across 8 medical datasets and 10 model families, tracking accuracy, hallucination/deviation rate, latency, throughput, bilingual performance, and prompt-following failures

Machine Learning Engineer *Apr 2022 – Jun 2024*

- Built multilingual medical data pipelines that transformed EHR-style context, clinical records, and medical QA sources into instruction-tuning datasets
- Automated data curation using PyTorch, OpenAI Batch API, and tool-chained LLM workflows to translate, restructure, and enrich records from 3 languages
- Designed synthetic data generation and quality-control workflows using seed generation, model-as-judge review, prompt iteration, and OpenAI chat-format standardization

Pegasystems **New York, NY**
Data Science Intern *Sep 2023 – Dec 2023*

- Reduced distributed ETL execution time by 50% through PySpark-based pipeline and partitioning optimizations
- Boosted data mapping accuracy to 93%+ by optimizing the fingerprint tagging and similarity scoring algorithms
- Increased workflow pattern detection by 72% through refined process mining strategies and event correlation across ERP/CRM logs

Pitney Bowes **New York, NY**
Data Science Intern *Jun 2021 – Dec 2021*

- Designed Snowflake SQL pipelines across 100+ parcel-scanning data schemas from U.S. facilities, improving operational visibility for stakeholders
- Achieved 98%+ anomaly detection accuracy and drove a 10% revenue increase by developing time-series regression models in Python using sliding-window backtesting for early alerting and process optimization
- Built Power BI dashboards with real-time refresh pipelines and dynamic DAX measures, enabling self-service operational analytics

EDUCATION

Columbia University, Fu Foundation School of Engineering & Applied Science **New York, NY**
MS in Data Science *Dec 2024*

CUNY Baruch College, Zicklin School of Business **New York, NY**
BBA in Computer Information Systems, Cum Laude with College Honors, Minor in Mathematics *Jul 2020*

TECHNICAL SKILLS & CERTIFICATION

Programming: Python, R, SQL, C++, HTML, CSS, JavaScript, UNIX Shell Scripting
Tools: Oracle/Snowflake, MongoDB, Alteryx, AWS, GCP, Azure, Hadoop, Spark, Tableau/Power BI, MS Office, TensorFlow
Certifications: AWS Certified Machine Learning Specialty (in progress)